

CHARLES NORTON

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EDUCATION

Grand Valley State University

Expected August 2027

Combined B.S./M.S.E. Degree Program

Master of Science in Engineering, Emphasis in Electrical and Computer Engineering

Bachelor of Science in Electrical Engineering

SKILLS

Programming Python (NumPy, SciPy, Matplotlib, SymPy), C, MATLAB, GNU Octave, Julia

Hardware STM32 Microcontroller (STM32F446RE), Soldering, Altium Designer

CAD SolidWorks, Fusion 360, 3D Printing

MILITARY EXPERIENCE

Embarkation Logistics Chief

Jun 2013 – Apr 2017

United States Marine Corps

Camp Lejeune, NC

- Trained in automated information systems for asset tracking, load planning, and force deployment visibility
- Directed a 500-person railhead operation during urgent movement operations

WORK EXPERIENCE

Controls Design Engineer

Aug 2025 – Dec 2025

Koops Automation Systems

Holland, MI

- Designed electrical schematics and controls documentation for industrial automation systems
- Independently developed controls design for a test box used to evaluate a medical drill system
- Contributed controls design for a machine that assembled an encased PCB, supporting one machine section from design through release

Engineering Co-op — Project Engineering

Jan 2025 – Apr 2025

Koops Automation Systems

Holland, MI

- Modified and debugged an artillery shell manufacturing system to support project progress
- Authored user documentation for machine interface procedures
- Prototyped a bagging system for car seats, contributing to customer interest in automation

Engineering Co-op — Controls Engineering

May 2024 – Aug 2024

Koops Automation Systems

Holland, MI

- Programmed Allen-Bradley PLCs and HMIs for a tractor seat manufacturing system
- Developed control logic for a three-station conveyor with integrated data tracking
- Supported system testing, troubleshooting, and commissioning

Hospitality and Safety Manager

Jan 2020 – Dec 2021

The B.O.B.

Grand Rapids, MI

- Managed a team of 25+ to maintain a safe, welcoming environment
- Maintained logs for assets, incidents, and employee reviews to support liability protection
- Enforced fire safety compliance based on city codes

PROJECTS

Line Following Robot: Fast autonomous robot designed to follow a marked path

Karaoke Machine: Audio system with adjustable gain, Bluetooth, and auxiliary input

Digital Monster Pet: Interactive embedded system simulating a responsive digital pet

Skeeball Trivia Game: Skeeball game integrated with trivia-based gameplay